

Loan Default Prediction

TEAM 25

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Problem Overview

01 Defaults create direct financial losses

When a borrower fails to repay, the lender loses principal, interest income, and recovery costs

02 Manual underwriting does not scale well

Large lending platforms need consistent and automated risk assessment across many applications

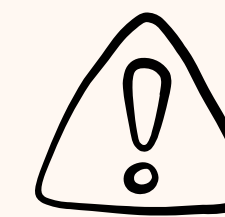
03 Risk reporting must be transparent

Managers need portfolio-level analytics to understand where default risk comes from

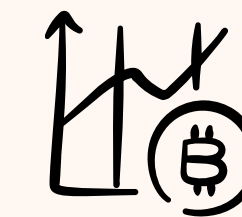
Default risk - Financial lost



Loan application



Default Risk Score



Business Action

Better risk prediction =
fewer missed risky loans
+ clearer portfolio reporting

Proposed Solution

01 HDFS / Hive

Distributed storage and warehouse layer

02 HiveQL EDA

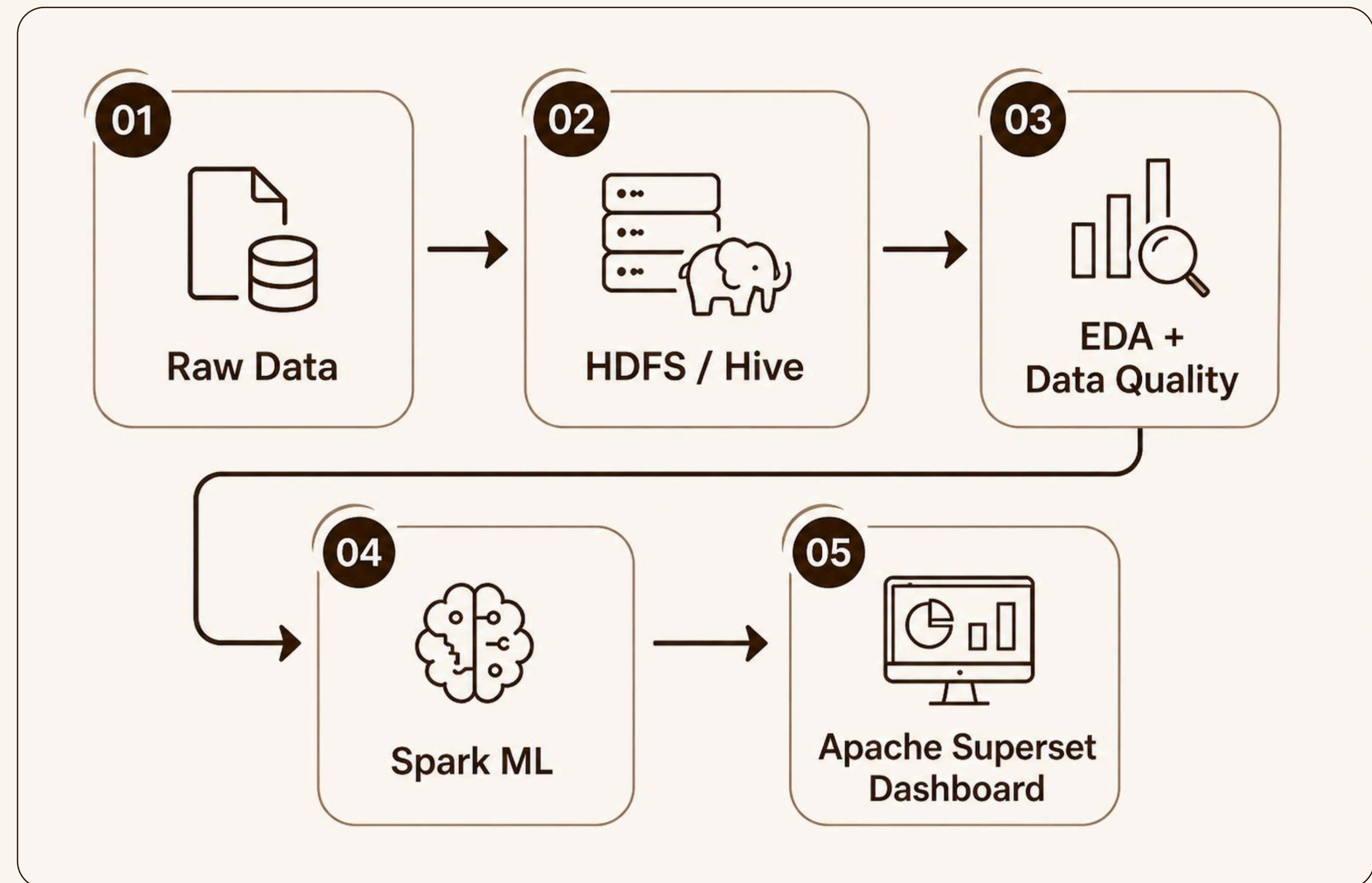
Default patterns and data quality checks

03 Spark ML

Distributed model training on YARN

04 Superset

Interactive dashboard for stakeholders

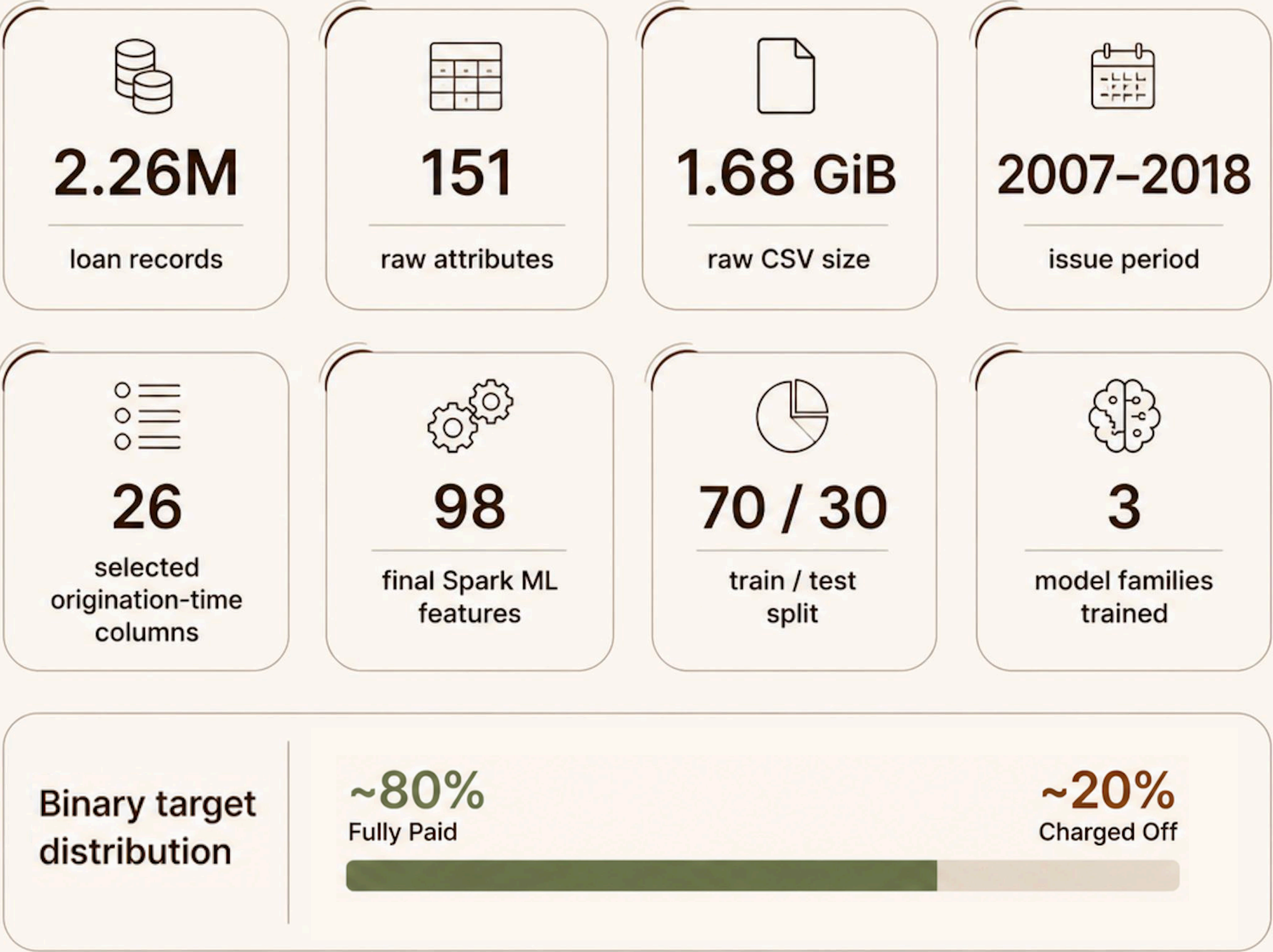


Dataset Characteristics

Lending Club Accepted Loans Dataset

A real-world collection of issued loans from Kaggle with borrower profiles, loan terms, credit-history indicators, and final repayment outcomes.

For modeling, we focus only on information available before or at loan issue time to mimic world-real scenario.



Stage I: Data Collection & Ingestion

01 Kaggle dataset

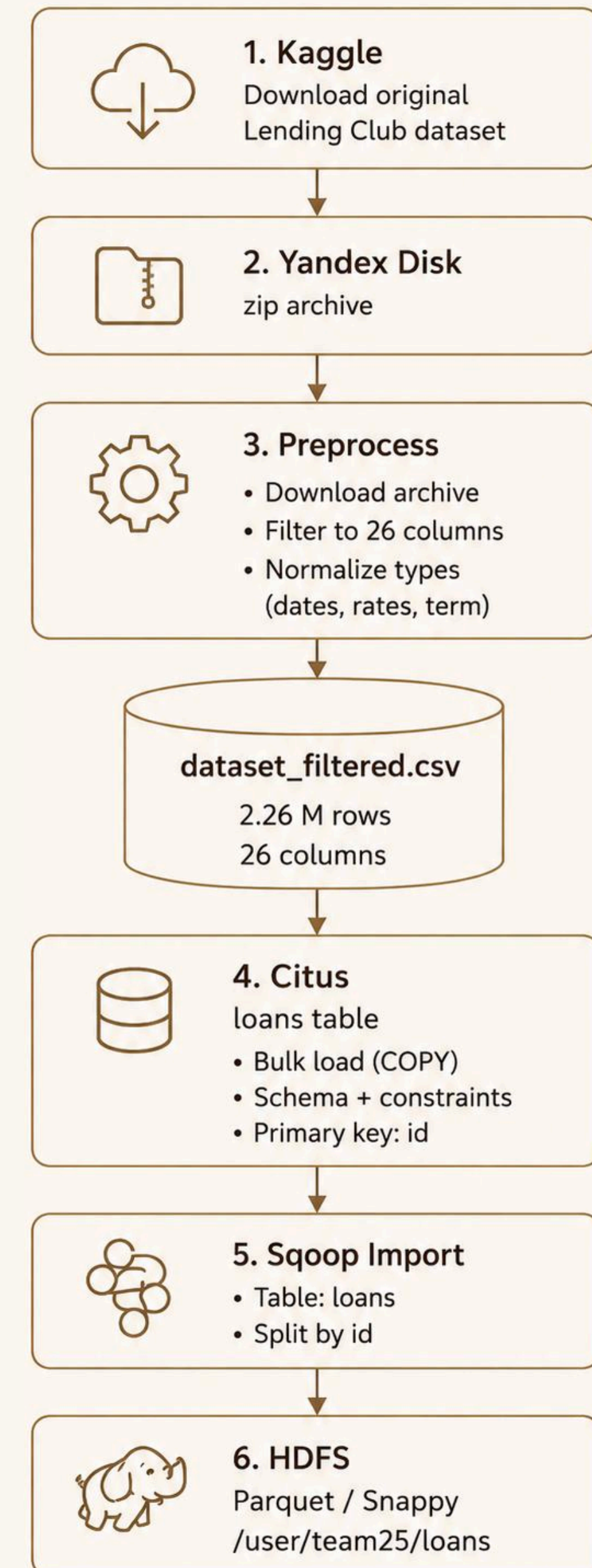
Initial data was loaded from Kaggle to Yandex Disk to eliminate possible changes of dataset

02 CSV normalization

CSV data was normalized to appropriate data types and loaded to Citus

03 Sqoop Export

Data from Citus was moved to HDFS via Sqoop for later usage in EDA and Modeling phases



Stage II: Hive Storage & Preparation

01 Hive tables

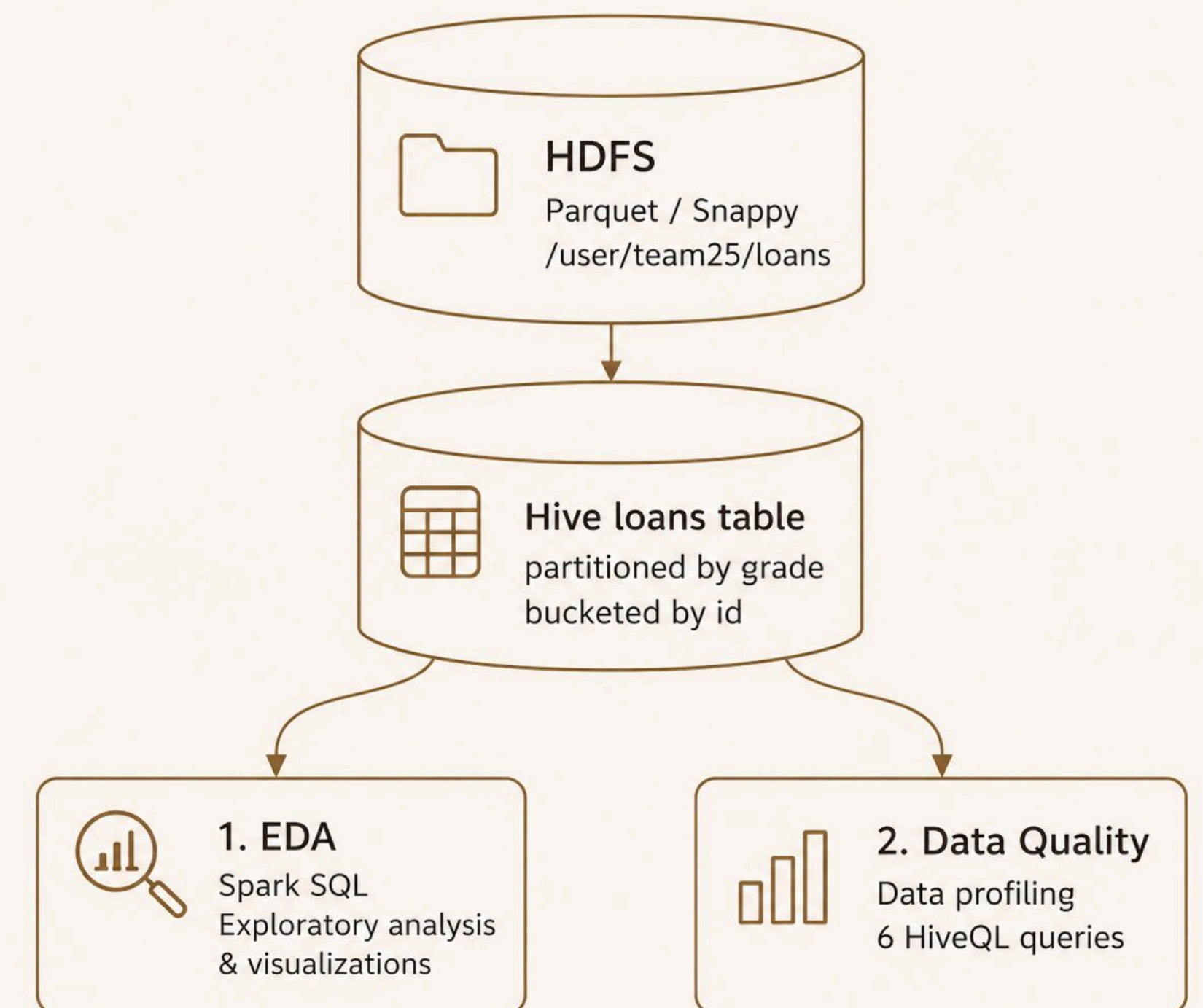
Raw Parquet output was loaded into a partitioned and bucketed Hive table, with timestamp type corrections applied

02 Data Quality

Dataset was profiled to identify anomalies, inconsistencies, and edge cases before running the full analysis.

03 EDA

Six HiveQL queries analyzed default patterns across grade, DTI, purpose, FICO score, and issue year to identify key risk signals



Stage II: EDA and Quality Checks

Analysis Performed

01 Default Risk by Loan Attributes

Analyzed how defaults correlates with features to find strongest approval-time risk signals

02 Temporal Trend Analysis

Tracked default rates and loan volumes by issue year to detect changes in borrower quality

03 Interaction Effects

Examined combined effects of features to find secondary risk patterns beyond individual once

04 Data Quality Checks

Identified and filtered invalid feature values according to standard banking rules

Essential insights

01 Prior Creditworthiness Predicts Default

Loan grade and FICO both show strong, monotonic correlation with default risk

02 Higher Debt Burden Means Higher Risk

Each 10-point Debt-to-Income increase adds ~4–5 percentage points to the default rate

03 Loan Purpose Strongly Predicts Default Risk

Default rates vary across purposes, from safest personal loans to high-risk business lending

04 Temporal instability

Loan volume and default rates varies significantly across issue years, with distinct patterns

Stage III: ML Data Preparation

01 Binary target

Fully Paid → 0, Charged Off → 1
Other loan statuses are excluded

02 Approval-time features only

We keep variables known before or at loan issue time
and remove post-origination leakage columns

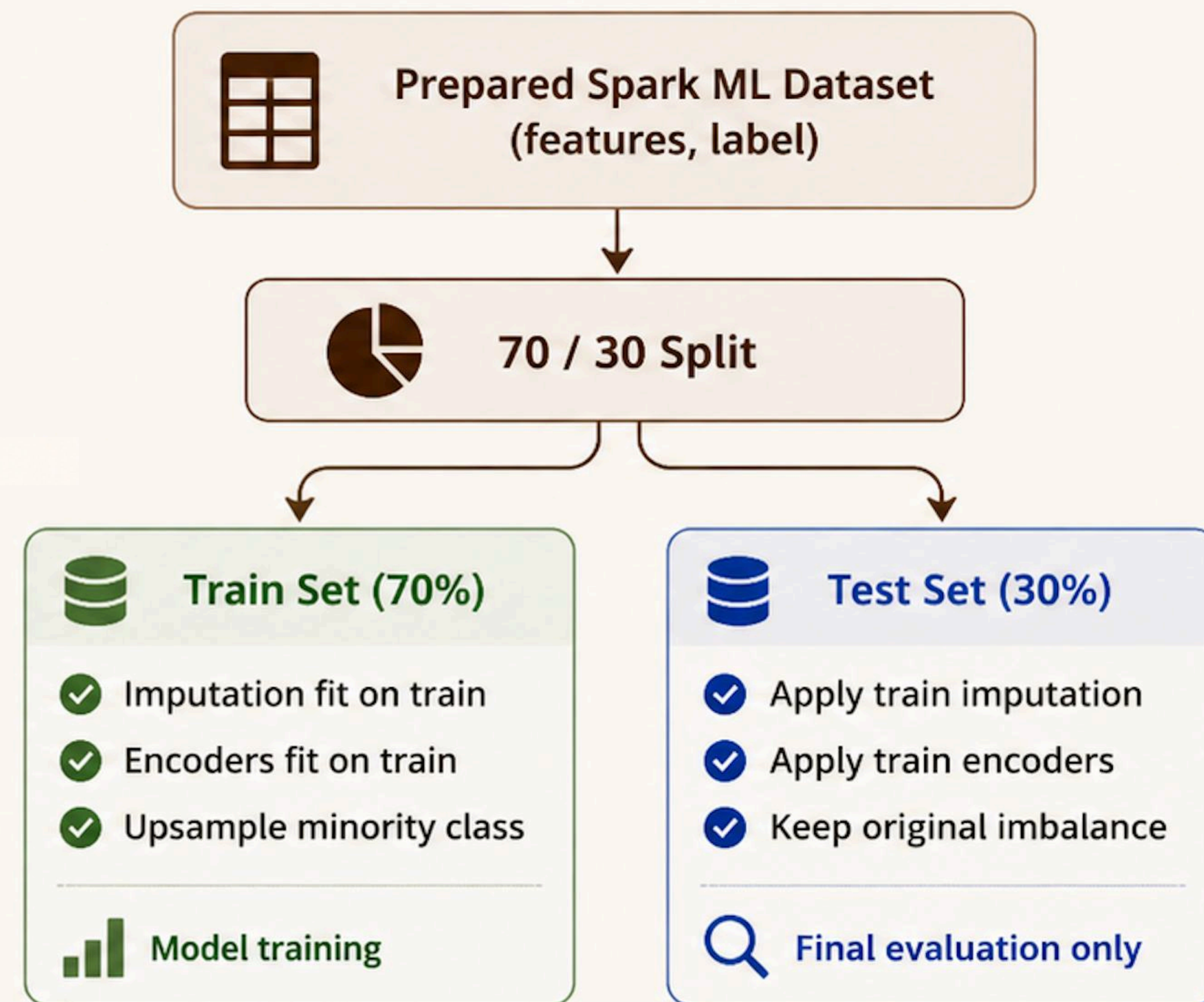
03 Temporal feature engineering

issue_year, cyclic *issue_month* encoding,
credit_history_months

04 Model-ready features

Combine numeric and categorical information
into a single 98-feature input for ML models

Train/Test Split



Stage III: Models and Tuning

Model families

01 Random Forest

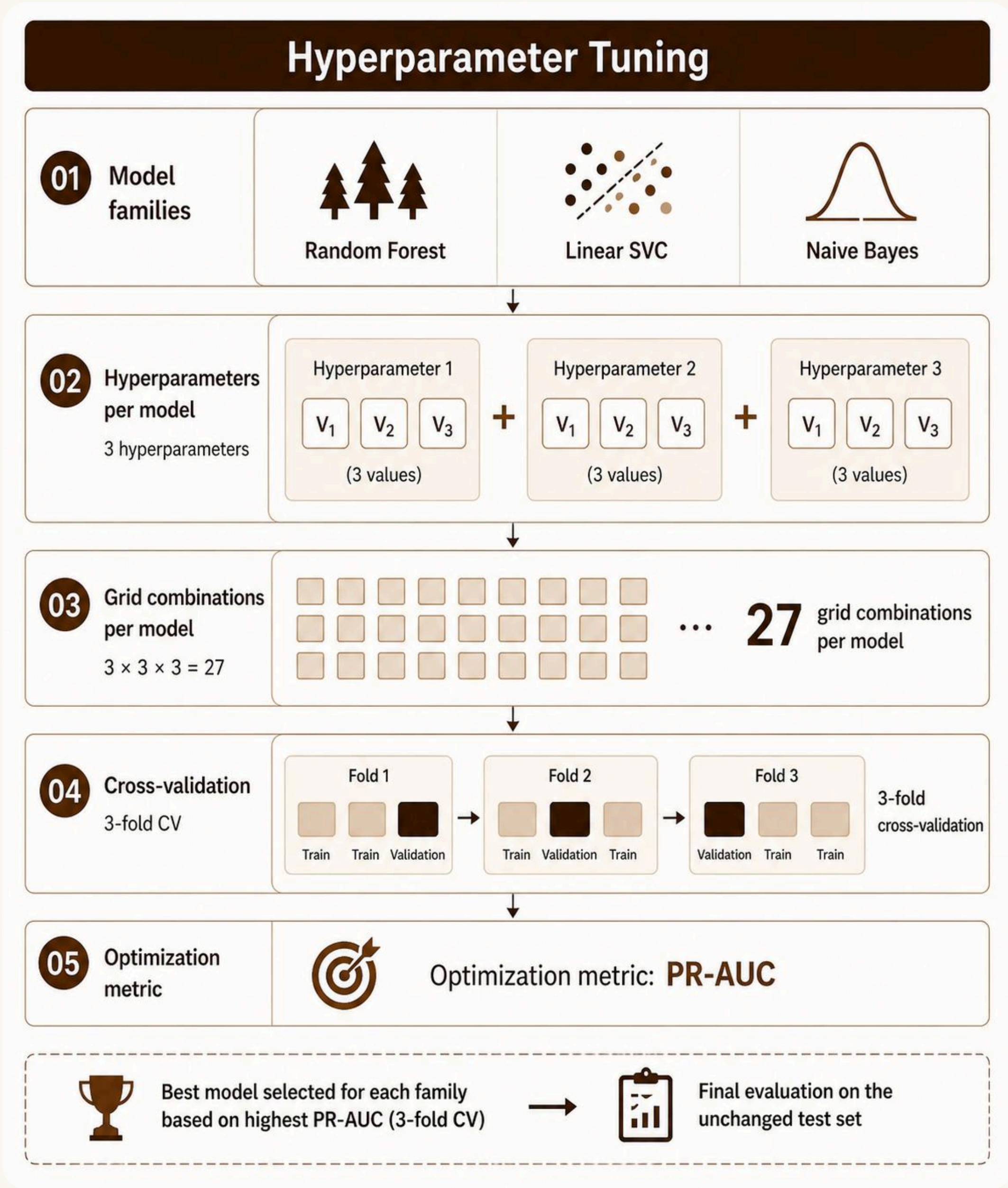
Tree-based ensemble for non-linear risk patterns

02 Linear SVC

Linear margin-based classifier

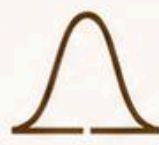
03 Naive Bayes

Probabilistic baseline with simple assumptions



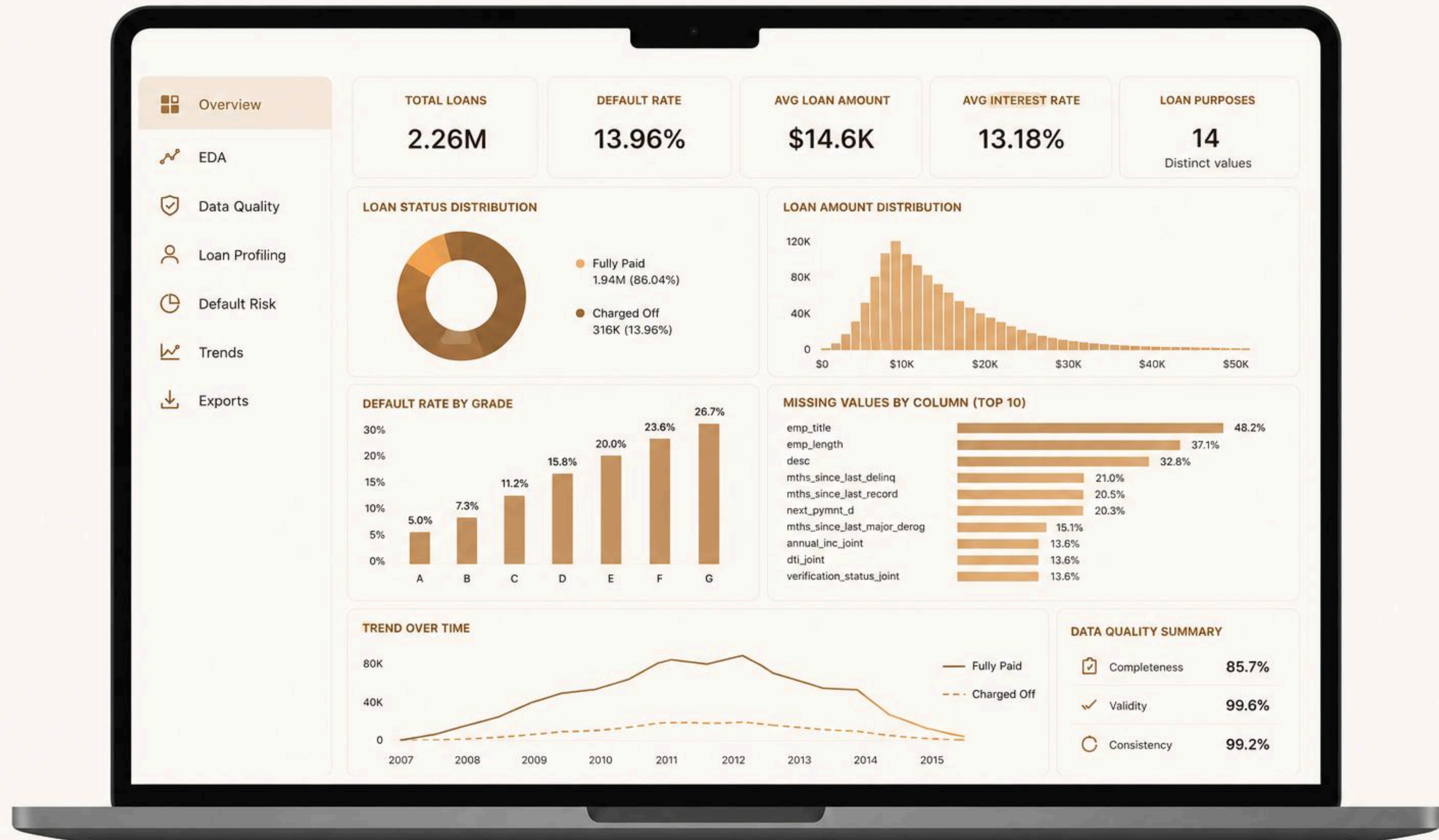
Stage III: Results

- Random Forest is the strongest tested model, but it should be used for risk screening, not automatic rejection.
- It detects ~71% of charged-off loans, but only ~30% of flagged risky loans are true defaults.
- Use case: prioritize applications for additional review.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
 Random Forest	0.6101	0.2991	0.7134	0.4215	0.7092	0.3696
 Linear SVC	0.6070	0.2939	0.6944	0.4130	0.6999	0.3574
 Naive Bayes	0.6174	0.2997	0.6892	0.4177	0.4583	0.1849

Dashboard Demo

Interactive dashboards for EDA and Data Quality



Challenges & Conclusion

Key challenges



DATE columns from Sqoop required manual conversion



Class imbalance caused many false positives



Leakage prevention required careful feature selection



Grid search was limited by YARN cluster resources



What we delivered



End-to-end reproducible Big Data pipeline



Hive EDA and data quality outputs



3 Spark ML model families trained on YARN



Superset dashboard for business reporting